

Hei Yu Chau

chauheiyu124@gmail.com | 07935 371713 | Swansea, UK | Full UK driving licence
linkedin.com/in/cheryl-chau-967840229 | github.com/Heyyouchao

PROFILE

First-Class Computer Science & Artificial Intelligence graduate with experience developing end-to-end AI applications, data processing pipelines and database-backed systems. Skilled in Python, SQL, machine learning, REST APIs, databases, Git and Docker through practical university projects. AWS Certified AI Practitioner with an interest in AI, data engineering and automation, using these technologies to solve practical problems.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, PHP, Java, Ada

Data & AI: Machine Learning, Scikit-Learn, PyTorch, TensorFlow, Feature Engineering, Data Preprocessing, Model Evaluation, Testing & Validation

Databases: MySQL, PostgreSQL, MongoDB

Development: REST APIs, Streamlit, Laravel, MVC Architecture, HTML/CSS, Object-Oriented Programming

Tools & Visualisation: Git, Docker, Jupyter Notebook, DBeaver, Plotly

Operating Systems: Windows, Linux (Ubuntu), macOS

PROJECTS

Final Year Project – Interactive Explainable AI for Cyber-Attack Detection in Power Systems

- Designed and developed an **end-to-end Python application** for detecting abnormal and potentially malicious behaviour in simulated smart-grid data.
- Built a **data processing and hierarchical machine-learning pipeline** covering preprocessing, feature engineering, anomaly detection, staged detection and model evaluation.
- Integrated backend processing and machine-learning outputs with a **Streamlit and Plotly dashboard** to visualise system status, IDS alerts, PMU waveforms and event logs for operator review and decision-making.
- Designed debug and live-style workflows to test individual cases, **validate model outputs and support human review**.

Cymru Hunt — Full-Stack Web Application (96%)

- Developed a **Laravel MVC** web application with **MySQL** backend and interactive map-based content system
- Designed and implemented **RESTful APIs**, guest and authenticated user workflows, and **role-based access control**
- Containerised the application using Docker** and developed database-backed features including comments, likes, activity tracking and events.

Traffic Light Controller (SPARK Ada) (92%)

- Developed a **formally verified** traffic control system ensuring safe signal-state transitions
- Applied defensive programming, validation and safety-focused system design principles to prevent invalid system states.

EDUCATION

Swansea University — BSc (Hons) Computer Science with Artificial Intelligence 2023 – 2026

First-Class Honours

Key Modules: Web Application Development (96%) | Big Data & ML (91%) | Critical Systems (90%)

CERTIFICATIONS

AWS Certified AI Practitioner (AIF-C01) — Amazon Web Services, 2026

EXPERIENCE

Student Ambassador — Swansea University June 2025 – Present

- Clearing Helpline Operator (Aug 2025 & Aug 2026):** Used internal database systems to guide applicants through Clearing and provide accurate information in a fast-paced environment.

PASS Leader — Swansea University Sep 2024 – Dec 2024

- Facilitated collaborative learning sessions, simplifying complex technical concepts through clear communication.